Constantine 2 University – Abdelhamid Mehri

Unit: DAW | Level: L2

Gym Management System

Final Project – Web Application Development

Project Overview

This project requires students to design and develop a fully functional web-based Gym Management System. Students must propose a gym brand name, logo, and visual theme. The system manages gym members, subscriptions, classes, trainers, and administrative functions.

Teams of up to three students from the same group are permitted. The project is divided into four progressive parts, each evaluated separately. Every part builds upon the previous, ensuring students develop both foundational and advanced web development skills.

Evaluation Schedule

Part	Topics Covered	Evaluation Period
Part 1 + Part 2	HTML Structure & Semantics & CSS Styling & Layout	Week of March 14
Part 3	JavaScript – Interactivity	Week of April 4
Part 4	PHP – Backend & Database	Week of April 25

General Project Rules

The following rules apply to all parts of the project:

- Teams must consist of up to 4 students, all from the same group.
- Each team must propose a unique gym brand name, logo concept, and color theme.
- Code must be well-organized, properly indented, and commented where necessary.
- Each part is evaluated independently: partial completion is graded.
- Academic integrity is strictly enforced. Plagiarism between groups will result in a zero for all involved parties.
- HTML and CSS must be valid (use the W3C validator). JavaScript must not produce console errors. PHP must run without warnings.
- All pages must be cross-browser compatible (tested in Chrome and Firefox).
- Students are responsible for any part of the code they have written.

Good luck to all students. Take pride in your work : this project is an opportunity to build a real-world portfolio piece.

Part 1: HTML Structure & Semantics

Duration: 2 Weeks | Technologies: HTML5 only

In this first part, students build the complete page structure of the Gym Management System using only HTML. The focus is on semantic markup, logical page organization, and content hierarchy. No styling or scripting is required at this stage : the goal is a clean, well-structured foundation that clearly communicates the content and layout intentions.

Client-Facing Pages

Page 1: Home Page (index.html) | Duration: Part 1	
Overview	The landing page of the gym website. Introduces the gym, displays key offerings, and guides visitors to explore more.
Structure	• Navigation bar with links to all main pages (Home, Classes, Membership, Trainers, Contact) • Hero section with gym name, tagline, and a call-to-action button (e.g., 'Join Now') • Section showcasing featured gym facilities (e.g., cardio room, weights area, pool) • Section listing membership plan highlights (Bronze, Silver, Gold) • Footer with contact info, social media links, and opening hours

HTML Focus	• Use semantic tags: <header>, <nav>, <main>, <section>, <article>, <footer> • Proper heading hierarchy: <h1> for gym name, <h2> for section titles • Use <ul>/<li> for navigation and feature lists • Anchor tags for navigation links and CTA button

Page 2: Classes Page (classes.html) | Duration: Part 1	
Overview	Displays all available gym classes with their schedules and basic details.
Structure	• Page header with title and short description • A table listing all classes: Class Name, Trainer, Day, Time, Duration, Difficulty Level • Each class entry links to a (placeholder) class detail page • Filter note section indicating upcoming JavaScript filtering (placeholder UI)
HTML Focus	• Structured <table> with <thead>, <tbody>, <tr>, <th>, <td> • Use <caption> for table accessibility • Use <a> tags within table cells for class detail links

Page 3: Membership Page (membership.html) | Duration: Part 1	
Overview	Presents the different membership plans and a registration form for new members.
Structure	• Three membership plan cards: Bronze (monthly), Silver (quarterly), Gold (annual) • Each card lists included benefits as a bullet list • Registration form: Full Name, Email, Phone, Date of Birth, Plan Selection (radio buttons), Terms checkbox, Submit button • Section with payment method icons (as placeholder images or text)
HTML Focus	• Use <form>, <fieldset>, <legend>, <label>, <input>, <select>, <textarea> • Use proper input types: text, email, tel, date, radio, checkbox • Use <dl>/<dt>/<dd> for plan feature descriptions

Page 4: Trainers Page (trainers.html) | Duration: Part 1	
Overview	Showcases the gym's professional trainers with their specializations.
Structure	• Grid of trainer profiles: photo (placeholder), name, specialty, years of experience, short bio • Each trainer links to a detail page (placeholder href) • Section with a 'Work With Us' recruitment note and link to contact page
HTML Focus	• Use <figure> and <figcaption> for trainer images • Use <article> for each trainer card • Use <address> for contact details

Page 5: Contact Page (contact.html) | Duration: Part 1	
Overview	Allows visitors to get in touch and find the gym's location details.
Structure	• Contact form: Name, Email, Subject, Message textarea, Send button • Gym address, phone number, and email in an <address> block • Placeholder section for an embedded map (using <iframe> placeholder or static text) • Opening hours listed in a table
HTML Focus	• Use <form> with proper labels and input types • Use <address> semantic tag for location data • Use <table> for opening hours schedule

Admin-Facing Pages (Basic Structure)

Page 6: Admin Login (admin-login.html) | Duration: Part 1	
Overview	A simple admin authentication page.
Structure	• Login form: Username field, Password field, Login button • Link to return to the main site • Placeholder for error message display
HTML Focus	• Use input type='password' for secure field • Form with method and action attributes (placeholder values)

Page 7: Admin Dashboard (admin-dashboard.html) | Duration: Part 1	
Overview	The main control panel for gym administrators.
Structure	• Sidebar navigation: links to Members, Classes, Subscriptions, Trainers, Orders • Summary stats section: Total Members, Active Subscriptions, Classes Today, Revenue This Month (static placeholder numbers) • Recent activity table: Last 5 new registrations with name, date, plan
HTML Focus	• Use <nav> with <ul> for sidebar • Use <table> for recent activity • Use semantic layout: <aside> for sidebar, <main> for content

Page 8: Product/Plan Management (admin-plans.html) | Duration: Part 1	
Overview	Admin page to view and manage membership plans.
Structure	• Table of existing plans with columns: Plan Name, Price, Duration, Features, Actions (Edit/Delete placeholders) • Add New Plan form: Plan Name, Price, Duration, Description, Submit button

HTML Focus	• Use <table> with placeholder action buttons • Use <form> with appropriate input types • Use <button type='button'> for action placeholders

Part 2: CSS Styling & Responsive Design

Duration: 2 Weeks | Technologies: HTML5 + CSS3

In Part 2, students apply CSS to style all pages built in Part 1. The focus is on creating a professional, visually consistent, and fully responsive interface. Students must define a gym brand identity through a cohesive color palette, typography, and layout system using Flexbox and CSS Grid.

Global Styling Requirements

Global: Brand Identity & Design System | Duration: Part 2	
Overview	Students must establish and apply a consistent visual language across all pages.
Colors	• Define a primary color (e.g., dark navy, black, or deep red), accent color, and neutral background • Apply colors consistently via CSS custom properties (variables): --primary, --accent, --bg, --text • Use distinct color treatment for client pages vs admin pages
Typography	• Import and apply a Google Font (e.g., Oswald for headings, Roboto for body) • Define font sizes using rem units for accessibility • Style headings (h1–h4), paragraphs, links, and button text consistently
Spacing & Layout	• Use a consistent spacing scale (e.g., 8px base unit: 8, 16, 24, 32, 48px) • Apply box-sizing: border-box globally • Use CSS resets or normalize.css to ensure cross-browser consistency

Client Pages Styling

Page 1: Home Page Styling | Duration: Part 2	
Overview	Transform the home page into an engaging landing experience.
Navigation	• Sticky navbar with logo on left, links on right using Flexbox • Hover effects on nav links (underline animation or color change) • Active page indicator (different color or underline on current page link)
Hero Section	• Full-width section with a background image or dark gradient overlay • Large centered text with the gym name and tagline • Styled CTA button with hover state (color change, box-shadow)
Facilities & Plans Sections	• CSS Grid layout: 3 columns on desktop, 1 column on mobile • Card-style components with border-radius, shadow, and padding • Icons or placeholder images with consistent sizing

Footer	• Dark background footer with three-column Flexbox layout • Responsive: stacks to single column on small screens

Pages 2–5: Other Client Pages Styling | Duration: Part 2	
Overview	Apply consistent branding to all remaining client-facing pages.
Classes Page	• Styled table with alternating row colors (striped effect) • Hover effect on table rows • Difficulty level indicators with colored badges (CSS only)
Membership Page	• Three-column plan card layout using Flexbox or Grid • Highlighted 'recommended' plan card with a distinct border or badge • Styled form with focus states, custom radio buttons and checkboxes
Trainers Page	• Responsive card grid: 3 columns desktop, 2 tablet, 1 mobile • Circular trainer profile photos using border-radius: 50% • Hover card effect (subtle lift using transform: translateY and box-shadow)
Contact Page	• Styled form fields with custom border and focus glow effect • Two-column layout: form on left, info block on right (desktop only)

Admin Pages Styling

Admin UI: Admin Dashboard Styling | Duration: Part 2	
Overview	Create a clean, functional admin interface distinct from the client site.
Layout	• Fixed sidebar navigation using Flexbox: sidebar on left, content area on right • Sidebar: dark background, white text, hover highlight on nav items • Content area: light gray background with white card containers
Stats Cards	• Four stat cards in a row using CSS Grid • Each card: white background, icon placeholder, large number, small label • Color-coded top border per card type (e.g., blue for members, green for revenue)
Tables	• Consistent admin table style: header row with dark background, striped rows • Action buttons (Edit, Delete) with distinct colors (blue/red) and hover states
Responsivenes s	• Sidebar collapses to top nav bar on mobile • Stat cards stack to 2x2 grid on tablet, 1 column on mobile

Responsive Design Requirements

All pages must be fully responsive across three breakpoints:

- Desktop: ≥ 1024px : full multi-column layouts
- Tablet: 768px – 1023px : adjusted grids (2 columns where applicable), collapsed nav
- Mobile: < 768px : single column, hamburger menu placeholder, touch-friendly tap targets (min 44px)

Use media queries with mobile-first approach. Flexbox and CSS Grid must both be demonstrated across different page sections.

Part 3: JavaScript – Interactivity & Dynamic UI

Duration: 3 Weeks | Technologies: HTML5 + CSS3 + Vanilla JavaScript (ES6+)

Part 3 introduces client-side scripting. Students will add dynamic behavior, form validation, DOM manipulation, and local data management using JavaScript. No external libraries (e.g., jQuery) are permitted : all code must use native ES6+ JavaScript.

Client-Side Features

Feature 1: Membership Registration & Validation | Duration: Part 3	
Overview	Enhance the membership registration form with real-time client-side validation.
Validation Rules	• Full Name: required, minimum 3 characters, letters only • Email: required, must match valid email format (regex) • Phone: required, must be a valid numeric format (8–15 digits) • Date of Birth: required, member must be at least 16 years old • Plan selection: at least one plan must be chosen • Terms checkbox: must be accepted before submission
UX Behavior	• Show inline error messages next to each invalid field on blur and on submit • Highlight invalid fields with a red border, valid fields with green border • Display a success confirmation message (or modal) on valid form submission • Prevent form submission if any field is invalid

Feature 2: Classes Filtering & Sorting | Duration: Part 3	
Overview	Make the classes table dynamic so users can filter and sort results without page reload.
Filtering	• Filter by Trainer: dropdown auto-populated from class data • Filter by Day of week: multi-select or button group • Filter by Difficulty: All / Beginner / Intermediate / Advanced buttons • Combined filters: all active filters apply simultaneously
Sorting	• Click on column headers (Class Name, Time, Duration) to sort ascending/descending • Show sort direction indicator (▲ / ▼) on active column
Implementation	• Store class data as a JavaScript array of objects • Render table rows dynamically using DOM manipulation (innerHTML or createElement) • Re-render table on every filter/sort change

Feature 3: Membership Cart & Plan Selection | Duration: Part 3	
Overview	Allow users to select a plan and review their choice before proceeding to checkout.
Cart Logic	• Users can click 'Select Plan' on a membership card to add it to their session • Only one plan can be selected at a time : selecting a new plan replaces the current one • Display a sticky mini-cart or summary bar showing the selected plan name and price • A 'Proceed to Register' button opens the registration form pre-filled with plan selection
Session Persistence	• Use sessionStorage to retain plan selection across page reloads • Clear selection on form submission or manual user action

Feature 4: Trainers Search & Modal Detail | Duration: Part 3	
Overview	Add search functionality and a detail modal for the trainers page.
Search	• Live search bar: filters trainer cards by name or specialty as user types • Display 'No trainers found' message if search returns empty results • Search is case-insensitive
Detail Modal	• Clicking a trainer card opens a modal with full trainer details: photo, name, specialty, bio, class schedule • Modal includes a close button (×) and closes on Escape key or outside click • Trainer data sourced from a JavaScript array of objects

Feature 5: Contact Form & Feedback | Duration: Part 3	
Overview	Validate the contact form and provide user feedback.
Validation	• Name: required, minimum 2 characters • Email: required, valid format • Subject: required, minimum 5 characters • Message: required, minimum 20 characters, character count display
Feedback	• Show toast notification on successful submission (auto-dismisses after 4 seconds) • Store submitted messages in localStorage for admin review (simulated backend)

Admin-Side JavaScript Features

Admin Feature 1: Simulated Member Management | Duration: Part 3	
Overview	Admin can view, add, edit, and remove members using JavaScript and localStorage.
CRUD Operations	• Display member list loaded from localStorage (pre-seeded with 5 sample members) • Add new member via a form: name, email, phone, plan, join date • Edit member details inline or via a modal form • Delete member with a confirmation dialog (confirm() or custom modal)
Search & Filter	• Live search: filter members by name or email • Filter by plan type: All, Bronze, Silver, Gold • Show total member count dynamically updated on filters

Admin Feature 2: Class Schedule Manager | Duration: Part 3	
Overview	Admin can manage the class timetable dynamically.
Features	• Add new class: name, trainer, day, time, duration, difficulty, capacity • Edit and delete classes from the schedule table • All changes persist in localStorage • Prevent duplicate class entries (same trainer, same day, same time)

Admin Feature 3: Dashboard Stats & Charts | Duration: Part 3	
Overview	Dynamically render dashboard statistics from stored data.
Dynamic Stats	• Calculate and display: total members, active subscriptions, classes per week, most popular plan • Stats update automatically when member or class data changes
Visual Chart (Optional)	• Render a simple bar chart using pure CSS or HTML5 Canvas (no libraries) • Chart shows member distribution by plan type

Part 4: PHP – Backend, Database & Full-Stack Integration

Duration: 3 Weeks | Technologies: HTML5 + CSS3 + JavaScript + PHP 8 + MySQL

In the final part, students replace the simulated JavaScript data with a real server-side backend using PHP and a MySQL database. The system becomes fully functional with persistent data storage, server-side validation, user authentication, and session management.

Database Design

Students must design and implement a normalized MySQL database with the following core tables:

- users : id, full_name, email, password_hash, phone, dob, role (member/admin), created_at
- memberships : id, user_id (FK), plan_id (FK), start_date, end_date, status (active/expired/cancelled)
- plans : id, name, price, duration_months, description, features
- trainers : id, name, email, specialty, bio, photo_path, years_experience
- classes : id, trainer_id (FK), name, day_of_week, start_time, duration_minutes, difficulty, capacity
- class_bookings : id, user_id (FK), class_id (FK), booked_at
- contact_messages : id, name, email, subject, message, submitted_at, read_status

Authentication System

Feature 1: User Registration & Login | Duration: Part 4	
Overview	Implement secure user authentication for both clients and administrators.
Registration	• Server-side validation: unique email check, password strength enforcement, age verification • Password hashed using PHP password_hash() before database insertion • Redirect to login page with success message on successful registration
Login	• Verify credentials using password_verify() • Create PHP session on success: store user ID, name, and role • Redirect: members go to their profile, admins go to dashboard • Show error message for invalid credentials without revealing which field is wrong
Session & Access Control	• Protect all admin pages with session check : redirect unauthorized users to login • Implement logout: destroy session and redirect to home page • Remember Me functionality using cookies (7-day token)

Client-Side Backend Integration

Feature 2: Dynamic Content from Database | Duration: Part 4	
Overview	Replace all hardcoded content with data fetched from the MySQL database.
Pages Updated	• Home Page: membership plan highlights fetched from plans table • Classes Page: class list fetched from classes JOIN trainers • Trainers Page: trainer cards fetched from trainers table • Membership Page: plan cards and prices fetched dynamically
Implementation	• Use PHP PDO with prepared statements for all database queries • Use include/require for reusable header, footer, and nav components • Pass data from PHP to JavaScript using json_encode() for client-side features

Feature 3: Member Profile & Subscription | Duration: Part 4	
Overview	Authenticated members can view and manage their profile and membership.
Profile Page	• Display member information pulled from the database • Edit profile form: update name, phone, and password (with current password confirmation) • Show current membership plan, start date, expiry date, and status
Class Bookings	• Members can book available classes from the classes page • View upcoming and past booked classes on the profile page • Cancel a booking (if class is more than 24 hours away)

Admin Backend Features

Admin Feature 1: Full Member Management | Duration: Part 4	
Overview	Admin manages all registered users through the database.
CRUD	• View all members with pagination (10 per page) • Add member manually (bypassing registration flow) • Edit any member's details and role • Delete or ban a member (set account status to 'banned')
Subscription Control	• View all active, expired, and cancelled subscriptions • Manually assign or extend a membership plan for a member • Filter members by plan type, registration date range, or status

Admin Feature 2: Plan & Class Management | Duration: Part 4

Overview	Admin can fully manage membership plans and gym classes via the database.
Plans	• Create, edit, and delete membership plans • Each plan: name, price, duration, feature list stored in database • Changes reflect immediately on the client membership page
Classes	• Create, edit, and delete class entries • View number of bookings per class and remaining capacity • Assign or reassign trainers to classes

Admin Feature 3: Contact Messages Management | Duration: Part 4	
Overview	Admin reviews and responds to messages submitted via the contact form.
Features	• View all submitted contact messages in a table (newest first) • Mark messages as Read / Unread • Delete messages from database • Display unread message count badge in admin sidebar

Admin Feature 4: Reporting & Data Export | Duration: Part 4	
Overview	Admin can view basic reports and export data.
Reports	• Revenue report: total income per month based on active subscriptions • Membership report: count of members per plan • Class popularity report: bookings per class sorted by demand
Export	• Export member list to CSV using PHP (fputcsv()) • Export subscription summary to a printable HTML page

Security Requirements

All backend features must implement the following security practices:

- Use PDO with prepared statements : never concatenate user input into SQL queries
- Sanitize all output with htmlspecialchars() to prevent XSS attacks
- Validate all form inputs on the server side, regardless of client-side validation
- Use CSRF tokens on all forms that modify data
- Store passwords using password_hash() / password_verify() : never store plaintext
- Restrict file upload types if profile photo upload is implemented (whitelist: jpg, png only)

Optional Enhancements (Bonus)

- Deploy the system to a free hosting service (e.g., InfinityFree, 000webhost, or Railway)
- Implement a REST API endpoint (e.g., /api/classes.php) that returns JSON for the JavaScript filter feature
- Add an email notification system using PHP mail() for membership confirmation
- Implement a QR code check-in system for class attendance tracking